

Add The File

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Concept: Implement a program that reads in a file and sums the numbers in the file.

1. Initiate variables and constants for data storage.
2. Get pathname variable and retrieve associated file
3. Place file into buffer
4. Iterate through buffer
5. First Loop: Iterate through second loop N number of times, retrieving N from the first line in the file
6. Second Loop: Iterate through file, adding each line to a total sum
7. Output final sum.

Code: Below is a copy-paste of my code.`section .data`

```
; get pathname
pathname dd 'randomInt100.txt'

; get size of buffer
bufferSize dd 1024

section .bss
; reserve a buffer with 1024 bytes
buffer: resb 1024

section .text

global _start

_start:

; open read to file
mov eax, 5
; take in pathname
mov ebx, pathname
; store int
```

```
mov ecx, 0
int 0x80
jmp read
```

read:

```
; move descriptor into eax
mov ebx, eax
; read in the file
mov eax, 3
; move the data from the read to the buffer
mov ecx, buffer
; move the length of the buffer into edx
mov edx, bufferSize
int 0x80
```

additionLoop:

```
; compare the current iter to counter
cmp rcx, buffer
; if equal, print
je print
; if not, add 1 to counter, go to getNumLoop
```

ext:

```
add rcx, 1
jmp getNumberLoop
```

getNumberLoop:

```
; compare current buffer to see if next char is newline
cmp byte [buffer + edi], 0xA
; if so, add buffer to ecx and go back to addition loop
add ecx, [buffer + edi]
je additionLoop
```

```

ext:
    add edi, 1
    jmp getNumberLoop

print:

    ; print the resulting int
    mov edx, eax
    mov eax, 4
    mov ebx, 1
    int 0x80

    ; exit
    mov eax, 1
    mov ebx, 0
    int 0x80

```

Output:

```

[lmagaha1@linux6 lab_filein]$ nasm -o main.o main.asm
[lmagaha1@linux6 lab_filein]$ ld main.o -o main
main.o: file not recognized: file format not recognized
[lmagaha1@linux6 lab_filein]$ █

```

Unfortunately my program is not fully functional. I have been having issues with compiling along with getting the full logic of my if-statements and loops implemented in Assembly. I plan to continue to work on and improve this assignment on my own until I can get it fully functional.